

NAME: Joel tan

ADMIN NUMBER:2302466D



SCHOOL OF INFORMATICS & IT

Diploma in Applied Artificial Intelligence AI and Ethics (CAI2C02)

Project

Student Name (Matric Number) : TAN JUN AN
JOEL(2302466D)

Tutorial Group : TC22

Tutor : Mr Gordon

Submission Date : 26/1/2025

Declaration of Originality

I am the originator of this work and I have appropriately acknowledged all other original sources used as my references for this work.

I understand that Plagiarism is the act of taking and using the whole or any part of another person's work, including work generated by AI, and presenting it as my own.

I understand that Plagiarism is an academic offence and if I am found to have committed or abetted the offence of plagiarism in relation to this submitted work, disciplinary action will

UNETHICAL USE OF AI

With the rapid growth of Artificial Intelligence (AI), AI has redefined how we approach and tackle real world problems across various sectors such as healthcare, transportation, entertainment, and law enforcement. AI is a very powerful tool that allows us to automate decision making, analyse huge amounts of data and improve our quality of life.

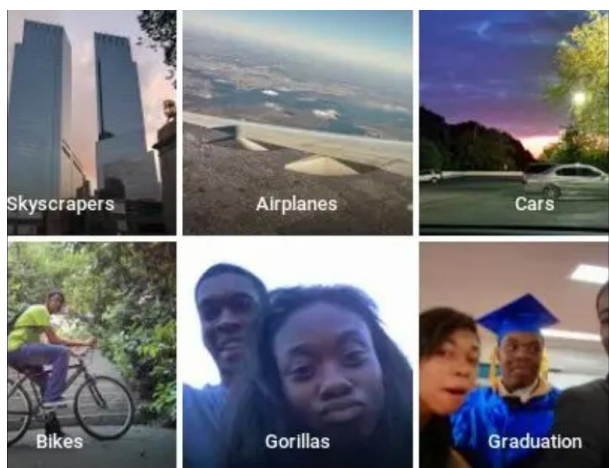
However, alongside its huge potential to revolutionize industries, AI could also pose significant ethical challenges. This is highly evident in applications such as facial recognition where issues such as privacy breach, biasness and lack of accountability become prominent. This essay carefully examines and analyses the inclusion of facial recognition, the ethical dilemmas it poses and the contributing factors to these challenges and possible solutions to solve the issue.

Introduction

What is the purpose of the Google AI Labelling App?

<https://www.bbc.com/news/technology-33347866>

The google labelling system taken from the article involves a photo-tagging feature or image labelling AI, the technology used in Google Photos primary purpose was to use and improve automated image categorization using AI. By automatically tagging photos with labels such as 'Skyscrapers', 'Airplanes', 'Cars', 'Bikes', 'Graduation'. The system aimed to allow users to categorise their images without manual effort which enhances user experience.



However, on 29 June 2015 A black couple faced a horrifying ethical failure when it mislabelled a photo of them as 'Gorillas'. This incident pointed out critical issues with AI biasness, which led to public backlash and calls for improved accountability and sensitivity in

AI systems. Such a fatal error damaged the reputation of google and highlighted the bigger implication of biasness in machine learning.

This catastrophic failure shows how the underlying datasets used to train the AI systems often lack diversity and variety which leads to biased or harmful outcomes. This incident triggered a discussion about the importance of inclusivity in AI training data. In response to the error, Google removed the ability of the system to label 'Gorillas'. This is a good temporary fix to address the problem. However, it does not address the root cause of the problem which is the training of the AI systems.

The case emphasizes the following below:

1. Bias Mitigation: Training datasets must have more variety and representative of diverse populations to avoid negative outcomes.
2. Transparency: Companies need to be open and aware about the risks and limitations of their AI systems and should only deploy it when they feel 100% confident.

What are facial recognitions mainly used for?

Facial recognition technology is a versatile tool that is utilized between various industries for various reasons which includes:

1. Security and Law Enforcement: It is made to identify suspects, monitor, and look after public spaces, also adds extra layer of security in areas such as airports.
2. Healthcare: It is used to identify patients for hassle-free check-ins and medical record access, and to monitor health conditions like strokes or emotional states through facial analysis.
3. Authentication: It is used to verify identity to secure financial transactions like buying an item online. It is also used to unlock devices like smartphones and laptops using facial recognition instead of passcodes.
4. Retail and Marketing: It is used to track customer behaviours so that the shop can improve its shopping experiences.
5. Social media and Entertainment: It is used to create augmented reality filters and effects on applications such as Instagram, snapchat.
6. Attendance and Monitoring: To track employee attendance in workplaces, libraries to only allow students to enter.

Ethical Issues of the AI implementation:

Racial and ethnic misidentification:

The google labelling system misinterpreted and miscategorized a black couple as 'Gorillas' which inherently highlights racial biasness in AI systems. Such biasness in AI implementations is very troublesome as they risk perpetuating negative and harmful stereotypes especially against communities that are the minority.

This not only undermines the fairness and inclusivity of AI systems but also raises serious ethical concerns when deployed in real-world applications.

Cultural sensitives and historical context:

<https://popularresistance.org/deep-racism-the-forgotten-history-of-human-zoos/>

The mislabelling incident has implied cultural and historical implications. It associates people of the African descent with such a term recalls a history of racism and systemic discrimination. In the 1950s, Africans and Native Americans were subjected to inhumane treatment. This was widely popular with events like in 1889, The World's Fair which was visited by 28 million people who lined up to see 400 indigenous people as the main attraction. These events represent a traumatic era for people of African descent and those that were involved.

That was a time that was tough and traumatic for people of the African descent and should be glad to have forgotten and moved on about it as we enter the 20th century. However, whether unintentional or due to the biased training data, the AI's error evoked painful reminders of historical injustices. Deploying technology without considering such historical and cultural risks would affect the affected communities.

Lack of accountability:

The labelling error have directly impacted the dignity of the two individuals involved. Being misidentified in such an offensive and demeaning manner which will make a person lose their sense of self respect and worth. Such harm leaves a very deep mark and cannot be undone easily, even with public apologies or corrections. AI errors like these can dehumanize individuals which causes emotional distress and losing their trust in technology.

When these errors occur it becomes unclear who is the main team responsible for the consequences, the developers deploying the AI system, or the AI itself. Ambiguity in responsibility, AI systems operate on very complex algorithms and vast datasets. When errors arise, it is difficult to trace back to who made the specific decisions so that certain punishments can be properly be dealt with.

Lack of transparency:

The labelling error demonstrated how opaque the decision-making processes of AI systems can be, both to the users and even to the developers themselves. Transparency in AI refers to how comprehensive and the clarity of how the AI system processes data, make decisions, and derive to outcomes.

Impacts of lack of transparency can lead to erosion of public trust, many users were questioning whether AI systems are reliable. Without transparency it is very challenging to convince the public to trust AI applications after experiencing AI failures themselves.

Lack of accountability:

<https://www.wired.com/story/when-it-comes-to-gorillas-google-photos-remains-blind/>

Without clear accountability, it becomes challenging and confusing to implement effective solutions to implement and come up with effective solution, it also sets a dangerous precedent where other organizations may not feel the need to rigorously test their AI systems before deployment. Lastly, affected individuals or users affected are less likely to trust AI system when AI organizations fail to take responsibility. Which gets us thinking whether big companies like google prioritize ethical considerations.

Clear accountability matters because it ensures that failures are properly addressed and fixed, organizations are incentivized to design and deploy AI systems ethically, affected parties have pathways for redress.

Insufficient error handling:

Error handling refers to the process and mechanism that are in place to detect, stop and mitigate errors in AI systems. How did the issues arise? Firstly, the system lacked pre-emptive checks done to the AI system to prevent any harmful or offensive classifications from being made, there were also no mechanism to prevent or stop results that could cause harm or be offensive towards a specific race or ethnicity.

The impacts of insufficient error handling:

1. The lack of insufficient error handling can lead to harm to affected individuals as the AI system resulted in an offensive and humiliating experience for the black couple that was involved.
2. For big organizations like Google, insufficient error handling can have a higher chance of facing public backlash and legal challenges.

Effective error handling mechanisms are crucial and critical for ensuring that AI systems operate ethically and reliably. They help to detect and stop harmful outcomes before it reaches out to the end users, they provide developers with actionable feedback to solve systemic issues and minimizes the risk of harmful misclassifications.

Possible factor(s) that may have contributed to these ethical issues.

Racial and ethnic misidentification,

Cultural Sensitivities and Historical context:

1. Biased and unfair training data, AI models are trained on datasets that refers to the data being trained having underlying patterns, which will then produce unfair, inaccurate data and be a disadvantage to certain social groups. The dataset may overrepresent certain demographics and underrepresent others, which may lead to skewed results.
2. Overgeneralization in Algorithm, AI systems may simplify complex facial features into rigid patterns, which makes them less accurate for individuals from diverse racial or ethnic backgrounds.
3. Insufficient variety in benchmark tests, AI performance is often tested using datasets that do not take into account for racial diversity, which may cause potential issues.
4. Historical Ignorance, the developer of the AI system can lack awareness of the cultural and historical implications of certain outputs, which causes the AI behaviour to be offensive and insensitive.

Lack of transparency:

1. Absence of auditing mechanisms, without audits from external sources companies have little incentive to disclose AI limitations or flaws.
2. Little or insufficient documentation, developers do not provide clear explanations about how the AI systems make decisions and how the errors made by AI are being handled.
3. Complexity of AI systems, AI systems are very complicated which makes it challenging to explain their behaviour so that people not in the industry can understand.

Lack of accountability:

1. Developers may over rely on automation, developers may trust the AI systems too much, which causes them to implement human oversight mechanisms to catch errors. In this case it would be (Human over the loop). Human oversight is necessary as AI models make predictions based on patterns in the dataset but lack contextual understanding which makes them more prone to errors. Without human checks, system may produce outputs that can be harmful.
2. Ambiguity in responsibility, different departments may be involved in the development of the AI system, which makes it not very clear who is ultimately responsible for when the AI system makes an error.
3. There aren't regulatory frameworks, the lack of laws and for governing AI accountability allows any organisations whether big or small to avoid consequences for ethical issues that may arise.

Insufficient error handling:

<https://www.geeksforgeeks.org/black-box-problem-in-ai/>

1. Insufficient and inadequate testing procedures, AI systems are often tested in very controlled environments, which fail to consider for real world events that has happened, which may result in unforeseen and unexpected errors.
2. There is pressure for rapid deployment, companies would prioritize speed-to-market over well checked AI system, which leaves the system more vulnerable to mistakes as less checks are being done.
3. Complexity of AI Algorithms, Deep learning models are treated as 'black boxes. Based on the article, black box in AI is a model that takes in an input, processes it and produces an output but does not provide a clear explanation or reason why the output was made, this includes convolutional neural networks (CNN) for image recognition therefore making it difficult for developers to find or anticipate for errors when training.

Solutions of the issues

Cultural Sensitivities and Historical context:

Proposed solution: Ethical review during development of the AI system

The developers should incorporate cultural and historical sensitivity checks as part of the AI system's training and designing process.

They should assemble a team of experts, which may include sociologists, ethicists to assess potential cultural sensitivity risks during the development. The developers should also try and use a more diverse dataset for training; this is to ensure that the AI system do not state or say any harmful stereotype or comments.

Advantages:

1. Prevents AI systems from offending specific groups unintentionally,
2. Promotes inclusivity and producing fair AI outputs

Disadvantages:

1. It takes a long time and a lot of effort to gather a variety input
2. Assembling a team of experts may costs a large sum of money and it is not easy to find these professionals.
3. The professionals may or may not guarantee to anticipate all potential cultural and historic sensitivities which means that the AI system might still produce unfair outputs.

Racial and Ethnic Misidentification:

Proposed solution: Using diverse training datasets and bias mitigation

The developers should use datasets that incorporate a balance and fair representation of racial and ethnic groups.

They could implement this by working with organisations or companies that specialises in gathering datasets to better reflect global diversity, regularly evaluating and updating training datasets to eliminate any existing biases.

Advantages:

1. Shows the public the commitment to try and be fair and an inclusive AI development team
2. Reduces the AI bias and increases system accuracy

Disadvantages:

1. Collecting and processing the datasets may require additional time and manpower
2. Risk of privacy concerns during data collections

Lack of Transparency:

<https://shap.readthedocs.io/en/latest/index.html>

Proposed solution: Interpretable AI model

The developers should use interpretable models and tools that give explanations for certain decisions the AI system make.

They could implement explainability tools like SHapley Additive ExPlanations(SHAP), to allows users to understand how different outputs are being made.

Advantages:

1. Able to understand the AI decision making processes

Disadvantages:

1. Interpretable models will sacrifice some accuracy and performance but in return we get to understand how different specific outputs are generated
2. Developing these explainability tools like SHAP can take up a lot of resources

Lack of accountability:

Proposed solution: Clearer Accountability Frameworks

The developers should establish accountability measures by defining roles of developers and organisations in the deployment of AI.

They could implement this by requiring organizations to maintain detailed documentation on the different model design, data usage and the steps that they follow, they could also introduce independent review boards to evaluate and test the AI systems for any potential risks and concerns.

Advantages:

1. Encourages other organisations to prioritise ethical AI practices and to be more aware of the public feedback.
2. Helps to allocate the blame fairly in case of any AI failure

Disadvantages:

1. Regulatory changes may be met with resistance from organisations
2. Increases the administrative burden

Insufficient Error handling:

Proposed solution: Enhanced error detection mechanisms

The developers should implement fail-safes and built in error detection in the AI system to catch harmful labels or outputs before being deployed into the real world. For example, during pre-processing of the data to notice and catch any sensitive terms or extreme cases which can reduce the chances of errors being reached to the end users after deployment.

They should regular audits of AI system for specific errors, also to implement human oversight during the image labelling to processes to verify and confirm the AI generated labels before being released to the public.

Advantages:

1. Reduces the number of occurrences of offensive and inappropriate outputs
2. Improves the public trust in AI systems

Disadvantages:

1. Time and cost will be increased for additional human intervention and technical advancements and improvements.
2. Potential delay in the deployment of the AI system due to more rigorous checks being done.

REFERENCES:

<https://shap.readthedocs.io/en/latest/index.html>

<https://www.geeksforgeeks.org/black-box-problem-in-ai/>

<https://www.wired.com/story/when-it-comes-to-gorillas-google-photos-remains-blind/>

<https://popularresistance.org/deep-racism-the-forgotten-history-of-human-zoos/>

<https://www.bbc.com/news/technology-33347866>